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United States Patent	12407975
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Ren; Zhang et al.

Coaxial speaker

Abstract

The present invention provides a coaxial speaker including a frame, a magnetic circuit system, a first vibration system and a second vibration system fixed on different sides of the magnetic circuit system. The first vibration system locates communicated with the second vibration system. The first vibration system includes a first diaphragm and a first voice coil. The second vibration system includes a second diaphragm and a second voice coil. The magnetic circuit system includes a first magnetic gap and a second magnetic gap. The first voice coil partially locates into the first magnetic gap, the second voice coil partially locates into the second magnetic gap. The first diaphragm is used for generating bass sound, the second diaphragm is used for generating treble sound. A sound hole is provided on the first diaphragm for emitting the treble sound. Bass sound and treble sound form a full-range speaker, which providing high-quality sound.

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Appl. No.:	17/913850
Filed (or PCT Filed):	July 20, 2022
PCT No.:	PCT/CN2022/106664
PCT Pub. No.:	WO2024/000685
PCT Pub. Date:	January 04, 2024

Prior Publication Data

Document Identifier	Publication Date
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Foreign Application Priority Data

CN

202221717092.1

Jun. 30, 2022

Publication Classification**Int. Cl.:** **H04R1/24** (20060101); **H04R7/12** (20060101); **H04R9/02** (20060101); **H04R9/04** (20060101); **H04R9/06** (20060101)**U.S. Cl.:****CPC** **H04R1/24** (20130101); **H04R7/127** (20130101); **H04R9/025** (20130101); **H04R9/045** (20130101); **H04R9/06** (20130101); H04R2400/11 (20130101)**Field of Classification Search****CPC:** H04R (9/025); H04R (7/16); H04R (1/24); H04R (9/06); H04R (9/063); H04R (9/02); H04R (2400/11); H04R (2499/11); H04R (9/045); H04R (7/18)**USPC:** 381/400-405; 381/409-412

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
12120495	12/2023	Ren	N/A	H04R 9/06
12155991	12/2023	Ren	N/A	H04R 9/046
2002/0168074	12/2001	Miyamoto	381/182	H04R 9/10
2013/0195293	12/2012	Ko	381/191	H04R 19/02
2021/0377668	12/2020	Park	N/A	H04R 1/44
2023/0292037	12/2022	Xu	N/A	H04R 1/24

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201781608	12/2010	CN	N/A

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Background/Summary**FIELD OF THE PRESENT INVENTION**

(1) The present invention relates to the field of electro-acoustic devices, and more particularly, to a coaxial speaker used in electronic speaker box products.

DESCRIPTION OF RELATED ART

(2) Coaxial speaker refers to the speaker with the integration of the treble unit and bass unit, and the treble unit and bass unit are set coaxially. The sound source of the treble unit and bass unit of the coaxial speaker can be located in front of the coaxial speaker, that is, the two speaker units sounding along a same direction. The sound source of the treble unit and bass unit of the coaxial speaker can also be located in front and back of the coaxial speaker respectively, that is, the two speaker units sounding along different directions.

(3) In related art, the treble unit and bass unit of the coaxial speaker usually each includes an independent vibration system and an independent magnetic circuit system, and the structure of the coaxial speaker is complex. The overall height of the coaxial speaker composed of the treble unit and the bass unit coaxially is increased, which leads to inconvenient installation due to the high height and reduces the applicability of the product. In addition, the separated treble magnetic circuit and bass magnetic circuit also increase the production cost, and reduce the market competitiveness of the product. In the related art, the effective area of the diaphragm in the speaker is limited, which cannot meet the requirements of the full sound range.

(4) Therefore, it is desired to provide a new coaxial speaker which can overcome the above problems.

SUMMARY

(5) In view of the above, the embodiment of the present invention provides a new coaxial speaker. By the present invention, the coaxial speaker has a compact structure and with high sound quality of the treble sound and bass sound.

(6) The present invention provides a coaxial speaker including: a frame, a magnetic circuit system mounted on the frame, a first vibration system fixed on one side of the magnetic circuit system and connected with the frame, a second vibration system fixed on the other side of the magnetic circuit system. The second vibration system locates coaxially with and communicated with the first vibration system. The first vibration system includes a first diaphragm and a first voice coil driving the first diaphragm to generate sounds. The second vibration system includes a second diaphragm and a second voice coil driving the second diaphragm to generate sounds. The magnetic circuit system includes a first magnetic gap and a second magnetic gap located outside of the first magnetic gap. The first voice coil at least partially locates into the first magnetic gap, and the second voice coil at least partially locates into the second magnetic gap. The first diaphragm is used for generating bass sound, and the second diaphragm is used for generating treble sound. A sound hole is provided on the first diaphragm for emitting the treble sound generated by the second diaphragm.

(7) As an improvement, the magnetic circuit system includes a magnetic bowl, the magnetic bowl including a side wall enclosing a channel and a bottom wall bending and extending from an end of the side wall closed to the first diaphragm, the magnetic circuit system further including a first magnet assembly disposed on one side of the magnetic bowl and a second magnet assembly disposed on the other side of the magnetic bowl, the first magnet assembly including a first magnet, the second magnet assembly including a second magnet and a third magnet located outside of the second magnet and spaced apart from the second magnet, the first magnetic gap formed between the first magnet, the second magnet, and the third magnet, the second magnetic gap formed between the second magnet and the side wall of the magnetic bowl.

(8) As an improvement, a through hole is provided on the first magnet for communicating the channel of the magnetic bowl and the sound hole.

(9) As an improvement, the second magnet assembly further includes a yoke supporting the second magnet and the third magnet, and the second diaphragm is mounted on a side of the yoke distal to the second magnet.

(10) As an improvement, the coaxial speaker further includes a support shelf, and the second diaphragm fixed with the yoke through the support shelf.

(11) As an improvement, the first vibration system further includes a voice coil support connecting the first diaphragm and the first voice coil, the first diaphragm including a first suspension, a first dome surrounding the first suspension, and a second suspension surrounding the first dome, the voice coil support connecting the first suspension, the first dome, and the second suspension to the first voice coil, an end of the first suspension mounted with the first magnet, an end of the second suspension mounted with the frame.

(12) As an improvement, the second diaphragm includes a second dome, a third suspension surrounding the second dome and the third suspension fixedly attached with the support shelf.

(13) As an improvement, the first magnet, the second magnet and the third magnet are magnetized along a vibration direction of the coaxial speaker, and the first magnet and second magnet are set opposite each other with a same pole, and the second magnet and third magnet are magnetized in opposite directions.

(14) As an improvement, the coaxial speaker further includes an elastic assembly elastically suspending the first vibration system, the voice coil support including a flat portion attached to the first dome, a first extending portion bending and extending from one end of the flat portion to the first voice coil and connected to the first voice coil, and a second extending portion bending and extending from the other end of the flat portion to the elastic assembly and connected to the elastic assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the exemplary embodiment can be better understood with reference to the following drawing. The components in the drawing are not necessarily drawn to scale, the emphasis instead being placed upon clearly illustrating the principles of the present invention. Moreover, in the drawings, like reference numerals designate corresponding parts throughout the several views.

(2) FIG. 1 is an illustrative isometric view of a coaxial speaker in accordance with one embodiment of the present invention.

(3) FIG. 2 is another illustrative isometric view of the coaxial speaker of FIG. 1.

(4) FIG. 3 is a partially exploded view of the coaxial speaker of FIG. 1.

(5) FIG. 4 is an exploded view of the coaxial speaker of FIG. 1.

(6) FIG. 5 is another exploded view of the coaxial speaker of FIG. 1.

(7) FIG. 6 is an illustrative cross-sectional view of the coaxial speaker taken along line A-A of FIG. 1.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

(8) The present invention will hereinafter be described in detail with reference to exemplary embodiments. To make the technical problems to be solved, technical solutions and beneficial effects of the present invention more apparent, the present invention is described in further detail together with the figures and the embodiments. It should be understood the specific embodiments described hereby is only to explain the disclosure, not intended to limit the disclosure.

(9) Referring to the FIGS. 1-6, the present invention provides one embodiment of a coaxial speaker **100**. The coaxial speaker **100** includes a frame **1**, a magnetic circuit system **2** mounted on the frame **1**, a first vibration system **3** fixed on one side of the magnetic circuit system **2** and connected with the frame **1**, a second vibration system **4** fixed on the other side of the magnetic circuit system **2**, an elastic assembly **5** elastically suspending the first vibration system **3**, and a support shelf **6** fixing the second vibration system **4** to the magnetic circuit system **2**. The first vibration system **3** locates coaxially with and communicated with the second vibration system **4**.

(10) The frame **1** is substantially rectangular, which can be a steel ring, and includes an upper

surface **11**, a lower surface **12** opposite to the upper surface **11**, and an outer surface **13** connecting upper surface **11** and the lower surface **12**. The first vibration system **3** is fixed to the upper surface **11** of the frame **1**, and the elastic assembly **5** is fixed to the lower surface **12** of the frame **1**.

(11) The magnetic circuit system **2** includes a magnetic bowl **21**, a first magnet assembly **22** disposed on one side of the magnetic bowl **21**, and a second magnet assembly **23** disposed on the other side of the magnetic bowl **21**. The first magnet assembly **22** locates coaxially with second magnet assembly **23**. The magnetic bowl **21** includes a side wall **211** enclosing a channel **210** and a bottom wall **212** bending and extending from an end of the side wall **211** closed to the first vibration system **3**. The first magnet assembly **22** includes a first magnet **221**. The bottom wall **212** of the magnetic bowl **21** supports the first magnet **221**. The first magnet **221** is an annular structure, and a through hole **222** is provided on a middle portion of the first magnet **221**. The second magnet assembly **23** includes a second magnet **231**, a third magnet **232** located outside of the second magnet **231** and spaced apart from the second magnet **231**, and a magnetic guiding plate **233** located on the third magnet **232**. The bottom wall **212** of the magnetic bowl **21** is stacked above the second magnet **231**. The first magnet **221** is disposed opposite to the second magnet **231**. The magnetic circuit system **2** further includes a yoke **24** supporting the second magnet **231** and the third magnet **232**. A first magnetic gap **25** is formed between the first magnet **221**, the second magnet **231**, and the third magnet **232**, the magnetic guiding plate **233**. A second magnetic gap **26** is formed between the second magnet **231** and the side wall **211** of the magnetic bowl **21**.

(12) The first magnet **221**, the second magnet **231** and the third magnet **232** are magnetized along a vibration direction of the coaxial speaker **100**, and the first magnet **221** and second magnet **231** are set opposite each other with a same pole, that is, the first magnet **221** and the second magnet **231** are magnetized in opposite directions. The second magnet **231** and third magnet **232** are magnetized in opposite directions. In present embodiment, the first magnet **221**, the second magnet **231**, the third magnet **232**, and the magnetic guiding plate **233** can all be hollow annular structures.

(13) The first vibration system **3** includes a first diaphragm **31**, a first voice coil **32** driving the first diaphragm **31** to generate sounds, and a voice coil support **33** connecting the first diaphragm **31** and the first voice coil **32**. The first voice coil **32** is at least partially located into the first magnetic gap **25**. The first diaphragm **31** is used for generating bass sound, and a sound direction of the first diaphragm **31** is upward along the vibration direction, that is, a sounding direction of the first diaphragm **31** is in front of the coaxial speaker **100**. The first diaphragm **31** includes a first suspension **311**, a first dome **312** surrounding the first suspension **311**, and a second suspension **313** surrounding the first dome **312**. The voice coil support **33** connects the first suspension **311**, the first dome **312**, and the second suspension **313** to the first voice coil **32**. An end of the first suspension **311** is mounted with the first magnet **221**, and the other end of the first suspension **311** is mounted with the voice coil support **33**.

(14) A recessing portion **241** is provided on the yoke **24** corresponding to a position of the first voice coil **32** to provide enough vibration space for the first voice coil **32**. A perforating hole **242** is provided passing through the yoke **24** to facilitate a communication between the first vibration system **3** and the second vibration system **4**.

(15) A sound hole **315** is provided on the first diaphragm **31** for emitting the treble sound generated by the second vibration system **4**. Specifically, the sound hole **315** is provided on the first suspension **311**. The first dome **312** is fixedly attached to the voice coil support **33**. One end of the second suspension **313** closed to the first dome **312** is mounted with the voice coil support **33**, and the other end is mounted with the frame **1**. Specifically, the other end of the second suspension **313** is attached to the upper surface **11** and extending to the outer surface **13** of the frame **1**.

(16) The second vibration system **4** includes a second diaphragm **41** and a second voice coil **42** driving the second diaphragm **41** to generate sounds. The second voice coil **42** is at least partially located into the second magnetic gap **26**. The second diaphragm **41** is mounted on a side of the yoke **24** distal to the second magnet **231**. The second diaphragm **41** is fixed to the yoke **24** through

the support shelf **6**. A height of the support shelf **6** should provide enough vibration space for the second diaphragm **41**. The second diaphragm **41** is used for generating treble sound, and the sound direction of the second diaphragm **41** is upward or downward along the vibration direction, that is, the sounding direction of the second diaphragm **41** is in front or back of the coaxial speaker **100**. When the second diaphragm **41** sounds in front of the coaxial speaker **100**, the treble sound is emitted by passing through the through the channel **210**, the through hole **222**, and the sound hole **315** in sequence. The second diaphragm **41** includes a second dome **411** and a third suspension **412** surrounding the second dome **411**. The third suspension **412** is attached to the support shelf **6**, and the support shelf **6** is a hollow annular structure.

(17) The elastic assembly **5** connects the voice coil support **33** and the frame **1**. The elastic assembly **5** includes a circuit board **51** and a support membrane **52** connected with the circuit board **51**. The first voice coil **32** is electrically connected to the circuit board **51**. Specifically, the coaxial speaker **100** further includes a terminal **7** connected to the circuit board **51**, and the first voice coil **32** is electrically connected to the circuit board **51** through the terminal **7**. In this embodiment, the number of the elastic assemblies **5** is two, and each elastic assembly **5** is arranged in a direction of a short axis of the coaxial speaker **100**. In other embodiments, the position of the elastic assembly **5** may be determined according to the situation.

(18) The voice coil support **33** includes a flat portion **331** attached to the first dome **312**, a first extending portion **332** bending and extending from one end of the flat portion **331** to the first voice coil **32** and connected to the first voice coil **32**, and a second extending portion **333** bending and extending from the other end of the flat portion **331** to the elastic assembly **5** and connected to the elastic assembly **5**.

(19) The present invention provides a coaxial speaker including: a frame, a magnetic circuit system mounted on the frame, a first vibration system fixed on one side of the magnetic circuit system and connected with the frame, a second vibration system fixed on the other side of the magnetic circuit system. The second vibration system locates coaxially with and communicated with the first vibration system. The first vibration system includes a first diaphragm and a first voice coil driving the first diaphragm to generate sounds. The second vibration system includes a second diaphragm and a second voice coil driving the second diaphragm to generate sounds. The magnetic circuit system includes a first magnetic gap and a second magnetic gap located outside of the first magnetic gap. The first voice coil at least partially locates into the first magnetic gap, and the second voice coil at least partially locates into the second magnetic gap. The first diaphragm is used for generating bass sound, and the second diaphragm is used for generating treble sound. A sound hole is provided on the first diaphragm for emitting the treble sound generated by the second diaphragm. The coaxial speaker in present invention arranges an annular bass vibration system for providing bass sound, a treble vibration system disposed coaxially with the bass vibration system for providing treble sound. Bass sound and treble sound form a full-range speaker, providing high-quality sound. Two voice coils using a common part of the magnetic system can further reduce the cost and improve the acoustic performance. Furthermore, an overall height of the coaxial speaker of the present invention is reduced.

(20) It is to be understood, however, that even though numerous characteristics and advantages of the present exemplary embodiment have been set forth in the foregoing description, together with details of the structures and functions of the embodiment, the disclosure is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms where the appended claims are expressed.

Claims

1. A coaxial speaker, comprising: a frame; a magnetic circuit system mounted on the frame, the magnetic circuit system comprising a first magnetic gap and a second magnetic gap, the first magnetic gap located outside of the second magnetic gap; a first vibration system fixed on one side of the magnetic circuit system and connected with the frame, the first vibration system comprising a first diaphragm and a first voice coil driving the first diaphragm to generate sounds, the first voice coil at least partially located into the first magnetic gap, the first diaphragm used for generating bass sound; and a second vibration system fixed on the other side of the magnetic circuit system, the second vibration system located coaxially with and communicated with the first vibration system, the second vibration system comprising a second diaphragm and a second voice coil driving the second diaphragm to generate sounds, the second voice coil at least partially located into the second magnetic gap, the second diaphragm used for generating treble sound; a sound hole provided on the first diaphragm for emitting the treble sound generated by the second diaphragm; wherein the magnetic circuit system comprises a magnetic bowl, the magnetic bowl comprising a side wall enclosing a channel and a bottom wall bending and extending from an end of the side wall closed to the first diaphragm, the magnetic circuit system further comprising a first magnet assembly disposed on one side of the magnetic bowl and a second magnet assembly disposed on the other side of the magnetic bowl, the first magnet assembly comprising a first magnet, the second magnet assembly comprising a second magnet and a third magnet located outside of the second magnet and spaced apart from the second magnet, the first magnetic gap formed between the first magnet, the second magnet, and the third magnet, the second magnetic gap formed between the second magnet and the side wall of the magnetic bowl.
 2. The coaxial speaker as described in claim 1, wherein a through hole is provided on the first magnet for communicating the channel of the magnetic bowl and the sound hole.
 3. The coaxial speaker as described in claim 1, wherein the second magnet assembly further comprises a yoke supporting the second magnet and the third magnet, and the second diaphragm is mounted on a side of the yoke distal to the second magnet.
 4. The coaxial speaker as described in claim 3, wherein the coaxial speaker further comprises a support shelf, and the second diaphragm fixed with the yoke through the support shelf.
 5. The coaxial speaker as described in claim 1, wherein the first vibration system further comprises a voice coil support connecting the first diaphragm and the first voice coil, the first diaphragm comprising a first suspension, a first dome surrounding the first suspension, and a second suspension surrounding the first dome, the voice coil support connecting the first suspension, the first dome, and the second suspension to the first voice coil, an end of the first suspension mounted with the first magnet, an end of the second suspension mounted with the frame.
 6. The coaxial speaker as described in claim 4, wherein the second diaphragm comprises a second dome and a third suspension surrounding the second dome, the third suspension fixedly attached with the support shelf.
 7. The coaxial speaker as described in claim 1, wherein the first magnet, the second magnet and the third magnet are magnetized along a vibration direction of the coaxial speaker, and the first magnet and second magnet are set opposite each other with a same pole, and the second magnet and third magnet are magnetized in opposite directions.
 8. The coaxial speaker as described in claim 5, wherein the coaxial speaker further comprises an elastic assembly elastically suspending the first vibration system, the voice coil support comprising a flat portion attached to the first dome, a first extending portion bending and extending from one end of the flat portion to the first voice coil and connected to the first voice coil, and a second extending portion bending and extending from the other end of the flat portion to the elastic assembly and connected to the elastic assembly.
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